CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 6: **We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of 102×15 is: 1

(a) 1515 **(b)** 1530 **(c)** 1540 **(d)** 1545

2. The value of 101×36 is: 1

(a) 3600 **(b)** 3672 **(c)** 3636 **(d)** 3646

3. The value of 68^2 is: 1

(a) 4692 **(b)** 4556 **(c)** 4624 **(d)** 4634

4. The value of $86^2 - 14^2$ is: 1

(a) 10000 **(b)** 7592 **(c)** 5184 **(d)** 7200

5. If $a = 3$ and $b = 5$, then $(a + b)^2$ equals: 1

(a) 34 **(b)** 4 **(c)** 64 **(d)** 49

6. $(a + b)^2$ equals: 1

(a) $a^2 + b^2$ **(b)** $a^2 + ab + b^2$ **(c)** $2a + 2b$ **(d)** $a^2 + 2ab + b^2$

7. $7 \times 8 = 8 \times 7$ illustrates the: 1

(a) commutative property **(b)** identity property
(c) distributive property **(d)** associative property

8. $(x + 3)(x + 4)$ equals: 1

(a) $x^2 + 12$ **(b)** $x^2 + 7x + 12$ **(c)** $x^2 + 7x + 7$ **(d)** $x^2 + 12x + 7$

9. The common mistake in $(a + b)^2 = a^2 + b^2$ is: 1

(a) extra a **(b)** missing the term $2ab$
(c) wrong sign **(d)** none of these

10. 101×99 equals: 1

(a) 10000 **(b)** 10001 **(c)** 9899 **(d)** 9999

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Factorise $a^2 - 49$. 2

12. Find 63×57 using $(60 + 3)(60 - 3)$. 2

13. Is $(a + b)^2 = a^2 + b^2$? Verify with $a = 2$, $b = 3$. 2

14. Write the commutative property with an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find 304^2 using a suitable identity. 3

16. Find 99^2 using a suitable identity. 3

17. Find 1002^2 using a suitable identity. 3

OR

Find 47^2 using a suitable identity. 3

18. Find 205^2 using a suitable identity. 3

OR

Find 998^2 using a suitable identity. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Expand $(2x + 3y)^2$. 5
(ii) Expand $(5a - 4b)^2$.
(iii) Find $x^2 + y^2$ if $x + y = 7$ and $xy = 12$.

OR

(i) Multiply $(x + 7)(x - 3)$ and verify at $x = 2$. 5
(ii) Find the product $(2a + 1)(a - 4)$.

20. **Case study:** A shopkeeper sells 98 notebooks at ₹ 24 each. 5
(i) Find the total using the distributive property.
(ii) If he sells 102 pens at ₹ 15, find that total.
(iii) Find the combined total.

OR

(i) Find $a^2 + b^2$ if $a - b = 5$ and $ab = 14$. 5
(ii) Find $(a + b)^2$.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND